



Microsoft ADC Cybersecurity Skilling Program

Lab 17 Assignment

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AZ-500

Lab 05: Securing Azure SQL Database

Exercise 1: Implement SQL Database security features

In this exercise, you will complete the following tasks:

Task 1: Deploy an Azure SQL Database

In this task, we use a template to deploy the lab infrastructure.

The screenshot displays the 'Edit template' page in the Microsoft Azure portal. The browser address bar shows the URL: <https://portal.azure.com/#view/HubsExtension/Temp...>. The page title is 'Edit template - Microsoft Azure'. A notification banner at the top right states 'Upload Completed for azuredeploy.json' with a file size of 2.14 KiB and the note '*Streaming upload*'. The main content area shows the Bicep template code for deploying an Azure SQL Database. The template includes parameters for the server name, administrator login, and password, and resources for the server, firewall rule, and database. The task pane on the right lists the following instructions:

1. Sign-in to the Azure portal
<https://portal.azure.com/>.
Note: Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab.
2. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Deploy a custom template** and press the **Enter** key.
3. On the **Custom deployment** blade, click the **Build your own template in the editor** option.
4. On the **Edit template** blade, click **Load file**, locate the **labfiles\Lab01\azuredeploy.json** file and click **Open**.
Note: Review the content of the template and note that it deploys an Azure SQL database.
5. On the **Edit template** blade, click **Save**.

The task pane also shows a progress bar for '23% Tasks Complete' and navigation buttons for 'Previous' and 'End'.

Jun 3 5:05 PM

Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

Microsoft.Template-20250603070201

Deployment

Deployment succeeded

Deployment 'Microsoft.Template-20250603070201' to resource group 'AZ500LAB11-lod51894525' was successful.

Pin to dashbo... Go to resource gr...

Your deployment is complete

Deployment name : Microsoft.Template-20250603070201

Subscription : AZ-500T00-A CSR 3

Resource group : AZ500LAB11-lod51894525

Start time : 6/3/2025, 7:02:06 AM

Correlation ID : f4ee5af2-8393-4428-b350-6088b6733a8d

Deployment details

Next steps

Go to resource group

Securing Azure SQL Database

39 Minutes Remaining

Instructions Resources Help

7. Click **Review + Create** and then click **Create**.

Note: Wait for the deployment to complete.

Task 2: Configure Advanced Data Protection

1. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Resource groups** and press the

52% Tasks Complete

< Previous End >

Task 2: Configure Advanced Data Protection

Jun 3 5:13 PM

Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

Server settings - Micro... Scan your Azure SQL... Azure Resource Mana...

https://learn.microsoft.com/en-us/azure/defender-for-cloud...

What are the express and classic configurations?

Configure vulnerability assessment for your SQL databases with either:

- Express configuration** – The default procedure that lets you configure vulnerability assessment without relying on external storage to store baseline and scan result data.
- Classic configuration** – The legacy procedure that requires managing an Azure storage account to store baseline and scan result data.

What's the difference between the express and classic configuration?

Configuration modes benefits and limitations comparison:

Securing Azure SQL Database

30 Minutes Remaining

Instructions Resources Help

2. On the **Resource groups** blade, in the list of resource group, click the **AZ500LAB11-lod51894525** entry.

3. On the **AZ500LAB11-lod51894525** blade, click the entry representing the newly created SQL Server.

4. On the SQL server blade, in the **Security** section, click **Microsoft Defender for Cloud**, select **Enable Microsoft Defender for SQL**.

5. On the SQL server blade, in the **Security** section, on the **Microsoft Defender for Cloud** page, in the **Microsoft Defender for SQL: Enabled at the subscription-level (Configure)** parameter, click **(configure)**.

6. On the **Server Settings** blade, review the information about pricing and the trial period, **VULNERABILITY ASSESSMENT SETTINGS** and **ADVANCED THREAT PROTECTION SETTINGS**.

7. Back to **Microsoft Defender for Cloud** blade, review **Recommendations** and **Security alerts**.

Task 3: Configure Data Classification

In this task, you will discover and classify information in

42% Tasks Complete

< Previous End >

Task 3: Configure Data Classification

In this task, we discover and classify information in SQL database for GDPR and data protection compliance.

Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

AZ500LabDb (az500112pcza66gi3kco) | SQL databases > AZ500LabDb (az500112pcza66gi3kco/AZ500LabDb)

SQL database

Save Discard Add classification Feedback

Currently database is using SQL Information Protection policy. Found 15 columns with classification recommendations →

Overview Classification Learn more - Getting Started Guide

15 columns with classification recommendations (Click to minimize)

Accept selected recommendations Dismiss selected recommendations Show dismissed recommendations

Column	Information type	Sensitivity label
SalesLT		
SalesOrderHeader		
TaxAmt		
Financial		
Confidential		

ENG US 7:19 AM 6/3/2025

Securing Azure SQL Database 24 Minutes Remaining 100%

Instructions Resources Help

Note: The classification engine scans your database for columns containing potentially sensitive data and provides a list of recommended column classifications.

5. Click the text message **We have found 15 columns with classification recommendations** displayed on blue bar at the top of the blade.

6. Review the listed columns and the recommended sensitivity label.

7. Enable the **Select all** checkbox and then click **Accept Selected Recommendations**.

Note: Alternatively, you could select only certain columns and dismiss others.

Note: You have the option to change the information type and sensitivity label.

8. Once you have completed your review click **Save**.

Note: This will complete the classification and 59% Tasks Complete

< Previous End >

Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

AZ500LabDb (az500112pcza66gi3kco) | SQL databases > AZ500LabDb (az500112pcza66gi3kco/AZ500LabDb)

SQL database

Save Discard Add classification Feedback

Currently database is using SQL Information Protection policy.

Overview Classification Learn more - Getting Started Guide

Successfully saved data classification

Successfully saved data classification for database AZ500LabDb.

Schema: 2 Table: 5 Filter by column na... Information typ... Sensitivity label...

Schema	Table	Column	Information type	Sensitivity label
dbo	ErrorLog	UserName	Credentials	Confidential

No classification recommendations

ENG US 7:22 AM 6/3/2025

Securing Azure SQL Database 21 Minutes Remaining 100%

Instructions Resources Help

sensitivity label.

7. Enable the **Select all** checkbox and then click **Accept Selected Recommendations**.

Note: Alternatively, you could select only certain columns and dismiss others.

Note: You have the option to change the information type and sensitivity label.

8. Once you have completed your review click **Save**.

Note: This will complete the classification and persistently label the database columns with the new classification metadata.

9. Back on the **Data Discovery & Classification** blade **Overview** tab, note that it has been updated to account for the latest classification information.

Task 4 : Configure auditing

In this task, you will first configure server level auditing and then configure database level auditing.

61% Tasks Complete

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Jun 3 5:23 PM

Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

AZ500LabDb (az500112pcza66gi3kco) | SQL databases > AZ500LabDb (az500112pcza66gi3kco/AZ500LabDb)

AZ500LabDb (az500112pcza66gi3kco/AZ500LabDb) |...

SQL database

Save X Discard + Add classification Feedback

Currently database is using SQL Information Protection policy.

Overview Classification Learn more - Getting Started Guide

SalesLT	Address	AddressLine1	Contact Info	Confidential
SalesLT	Address	AddressLine2	Contact Info	Confidential
SalesLT	Address	City	Contact Info	Confidential

No classification recommendations

63% Tasks Complete

Task 4 : Configure auditing

In this task, you will first configure server level auditing and then configure database level auditing.

63% Tasks Complete

< Previous End >

Task 4 : Configure auditing

In this task, we first configure server level auditing and then configure database level auditing.

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Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

Home > AZ500LAB11-lod51894525 > az500112pcza66gi3kco

az500112pcza66gi3kco | Auditing

SQL server

Save X Discard

Feedback

Azure SQL Auditing

Azure SQL Auditing tracks database events and writes them to an audit log in your Azure Storage account, Log Analytics workspace or Event Hub.

Learn more about Azure SQL Auditing

Enable Azure SQL Auditing

Audit log destination (choose at least one):

☒ Storage

Subscription *

Deployment succeeded

Deployment 'Microsoft.StorageAccount-20250603072756' to resource group 'AZ500LAB11-lod51894525' was successful.

Go to resource Go to resource group

6. Choose your **Subscription** from the dropdown list.

7. Click **Storage account** and choose **Create new**.

7. Click **Create New** and then on the **Create storage account** blade, in the **Name** box, type a globally unique name consisting of between 3 and 24 lower case letters and digits, click **OK**.

Note: You may need to refresh the browser before the storage account becomes available.

8. Back on the **Auditing** blade, under **Advanced properties** set **Retention (days)** to 5.

9. On the Auditing blade, click **Save** to save the auditing settings.

Note: If you receive an error message regarding invalid storage container path then the storage account may not have been provisioned yet. Wait a few minutes, click **Storage account**, on the **Choose storage account** blade, select the newly created storage account and back on the Auditing blade, click **Save**.

10. On the server blade, in the **Settings** section, click **SQL Databases**.

76% Tasks Complete

< Previous End >

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Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

Microsoft Azure

Home > az500112pcza66gi3kco

az500112pcza66gi3kco | Auditing

SQL server

Save X Discard

Feedback

Azure SQL Auditing

Azure SQL Auditing tracks database events and writes them to an audit log in your Azure Storage account, Log Analytics workspace or Event Hub.

[Learn more about Azure SQL Auditing](#)

Enable Azure SQL Auditing ☒

Audit log destination (choose at least one):

☒ Storage

Subscription *

Successfully saved server Auditing settings

Successfully saved Auditing settings for server: az500112pcza66gi3kco.

Securing Azure SQL Database

13 Minutes Remaining

Instructions Resources Help

3. Choose your **Subscription** from the dropdown list.

6. Click **Storage account** and choose **Create new**.

7. Click **Create New** and then on the **Create storage account** blade, in the **Name** box, type a globally unique name consisting of between 3 and 24 lower case letters and digits, click **OK**.

Note: You may need to refresh the browser before the storage account becomes available.

8. Back on the **Auditing** blade, under **Advanced properties** set **Retention (days)** to 5.

9. On the Auditing blade, click **Save** to save the auditing settings.

Note: If you receive an error message regarding invalid storage container path then the storage account may not have been provisioned yet. Wait a few minutes, click **Storage account**, on the **Choose storage account** blade, select the newly created storage account and back on the Auditing blade, click **Save**.

10. On the server blade, in the **Settings** section, click **SQL Databases**.

78% Tasks Complete

< Previous End >

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Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

Microsoft Azure

Welcome to SQL Database Query Editor

SQL server authentication

Login *

sqladmin

Password *

Cannot open server

'az500112pcza66gi3kco' requested by the login. Client with IP address '185.254.59.120' is not allowed to access the server. To enable access, use the Azure Management Portal or run sp_set_firewall_rule on the master database to create a firewall rule for this IP address or address range. It may take up to five minutes for this change to take effect.

[Allowlist IP 185.254.59.120 on server](#)

Securing Azure SQL Database

54 Minutes Remaining

Instructions Resources Help

13. On your SQL database Overview page in the Azure portal, select **Query editor (preview)** from the left menu. Try to sign in, you might fail on password, firewall rule for your IP address, everything gets audited. Try successful login as well, run query and you might find more details in audit logs

14. switch back to DB, Auditing and Click **View Audit Logs**.

15. On the **Audit records** blade, note that you can switch between Server audit and Database audit.

Results: You have created a SQL server and database, configured data classification, and auditing.

Clean up resources

Remember to remove any newly created Azure

88% Tasks Complete

< Previous End >

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Securing Azure SQL Database — Mozilla Firefox

https://labclient.labondemand.com/LabClient/20c08939-4ca6-4241-8881-2bf90733fea7

Audit records x Ledger docu x Scan your Az x Azure Resour x +

https://portal.azure.com/#view/SqAzureExtension/E...

Microsoft Azure Search resources, services, and docs (G+/)

Refresh Filter Log Analytics View dashboard

This blade provides a sample of audit logs with limited fields within 1 hour into the past from the selected End-Time (which is ...)

Switch to Bash Restart Manage files New session Editor

MOID: Azure Cloud Shell now includes Predictive IntelliSense! Learn more: <https://aka.ms/CloudShell/IntelliSense>

VERBOSE: Authenticating to Azure ...

VERBOSE: Building your Azure drive ...

PS /home/labuser-51894525> Remove-AzResourceGroup -Name "AZ500LAB11-lod51894525" -Force -AsJob

Id	Name	PSJobTypeName	State	HasMoreData	Location	Command
1	Long Running O...	AzureLongRunni...	Running	True	localhost	Remo...

PS /home/labuser-51894525>

ENG US 7:41 AM 6/3/2025

Securing Azure SQL Database 47 Minutes Remaining 100%

Instructions Resources Help

resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.

1. In the Azure portal, open the Cloud Shell by clicking the first icon in the top right of the Azure Portal. If prompted, click **PowerShell** and **Create storage**.
2. Ensure **PowerShell** is selected in the drop-down menu in the upper-left corner of the Cloud Shell pane.
3. In the PowerShell session within the Cloud Shell pane, run the following to remove the resource group you created in this lab:


```
powerShell
Remove-AzResourceGroup -Name "AZ500
```
4. Close the Cloud Shell pane.

Congratulations

You have successfully completed this lab. Click **End** to mark the lab as **Complete**.

100% Tasks Complete

< Previous End >

Summary

In this task we learn:

- Protection against attacks such as SQL injection and data exfiltration.
- Ability to discover and classify database information into categories such as Confidential.
- Ability to audit database server and database queries and log events.